

Lift Indications

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Represented here is a project on lifts' indications. It consists of Raspberry Pi (Raspi), MCP23008 8-bit I/O expander with serial interface, MCP23017 16-bit I/O expander with serial interface, DS18B20 digital thermometer, an alphanumeric 16 × 4 LCD and a few other components.

Many a time, we, in the industry, limit ourselves due to the overwhelming nature of proprietary items, combined with lack of money, to renovate and modernise old equip-

ment. Even though it is no longer under warranty, we are still afraid to carry out the maintenance work ourselves and often wait for the OEM to do it for us. We usually end up paying a hefty fee, too.

One such item is a boiler lift. We have quite a large number of these behemoths that still work but keep breaking down, requiring maintenance and eventual replacement over the years. One of the many problems with these lifts is that for indications, these use +24V DC, which lights small bulbs and on a number of occasions the bulbs fuse. On top of this,

these special long-glass incandescent bulbs are not easily available in the market.

To solve this problem, we tried using LEDs but these cause other problems. Even when there is no indication, due to the threshold limit, voltages present in the indication line and the LEDs never go off—so these keep glowing.

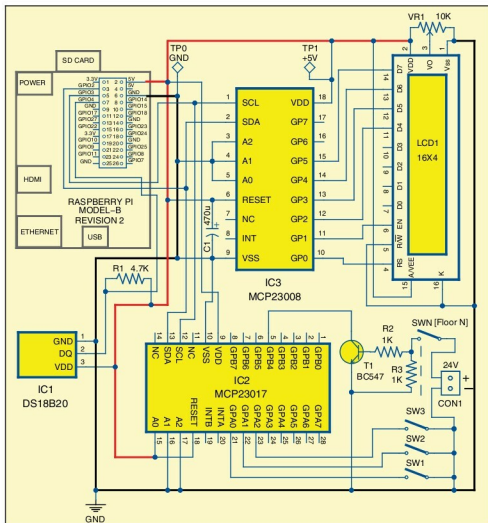
When my electrical maintenance guy expressed complete helplessness over this problem, I talked to him and after some trial and error, came out with an elegant solution. At the heart of the solution

lies Raspi computer, and the lift cage indications are completely replaced by a sophisticated 16 × 4 LCD panel, which shows temperature, clock, floor number and even makes audio announcement for floor numbers like the ones we find in modern Otis or Kone capsule lifts.

Raspi in action

We tried to build this project on Arduino but due to Arduino's limitations on audio voice reproduction, we shifted to Raspi. However, due to limitations of GPIOs on Raspi, we tied up one MCP23008 and MCP23017 each on I2C bus, so that the MCP23008 is used for running the 16 × 4 character LCD display for display purpose and the MCP23017 is used for receiving input signals from the several landings floors.

MCP23017 has 16 GPIOs, which can all be used for switching; it can provide indications for lift systems with up to 16 floors. Add one more MCP23017 and you can add another 16 floors.



Circuit diagram for lift indications

